

Brain Regions as Difference-Makers*

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Abstract

It is common to speak of brain regions for particular cognitive functions: regions for reading, for seeing faces, or for doing math. This suggests that brain regions have a single function which they always and uniquely perform. Advances in neuroscience show that this simple picture cannot be correct. I suggest that we ought to instead read ‘region for’ as designating brain areas that *make a difference* to the performance of personal-level activities. I discuss how brain regions can have specific and systematic relationships to personal-level activities, and argue that this combination allows neuroimaging constrain cognitive theories indirectly.

1 Introduction

1.1 Regions-for

We often hear talk of brain regions *for* this or that cognitive process. The popular press delivers up brain regions for reading, or religious fervor, or for loving your wife. Scientists are usually more cautious. They speak of regions for strictly *cognitive*-level functions rather than personal-level tasks—so for the processing that underlies reading, rather than for reading *per se*. Nevertheless, the locution—that some regions are *regions-for* something or other—is preserved.

What does it mean to say that R is the region for f ? I think most people who use the locution have a particular reading in mind: To say that R is the region for f is to say that (a) R performs f , (b) R performs only f , and (c) only R performs f . So a brain region for reading, say, is *specialized for reading*: it does all of the distinctively reading-related things, it can’t do anything else, and if something happens to R you’re done as far as reading goes. Call this the *standard reading* of the locution.

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The standard reading fits well with a certain model of brain function—inspired by Fodor, though no longer endorsed by him—on which the scientifically tractable bits of the brain form a kind of cognitive assembly line. Raw information comes in at the periphery. That information gets successively transformed by a series of specialized workstations. Those specialized workstations do only one thing, do it well, and do it largely without interference. When they’re done, they plonk their product onto the neural conveyor belt, and it heads over to the next station. To say that R is the region for f is to identify the specialized work that R does in the assembly line.

I’m interested in the standard reading because I’m interested in neuroimaging. I care particularly about how we might use neuroimaging to choose between competing cognitive theories. The standard reading usually plays a role in attempts to use neuroimaging in this way. But I think it’s problematic—so problematic, in fact, that we ought to ditch it and try something new. To motivate that, I’ll begin by discussing problems with the standard reading.

1.2 Problems with the Standard Reading

The standard reading faces several problems. I’ll focus on three: two metaphysical and one methodological. This paper is largely devoted to solving the two metaphysical problems. That solution is partly motivated, however, by its potential to deal with the methodological problem. So I’ll introduce all three, solve the metaphysical problems, and come back around in the final section to say something about methodology.

1.2.1 Nonspecificity

The standard story claims that brain regions do one thing and only one thing. At a suitable level of characterization, then, there should be a one-one mapping between cognitive functions and brain regions. There is now considerable evidence that this is not the case; the mapping is many-many. So brain regions are not specific to cognitive functions.

In the one direction, each brain region contributes to many distinct cognitive functions. The posterior lateral fusiform gyrus (PLF), for example, was originally identified as the ‘Visual Word Form Area’ on the basis of differential activity in word-reading experiments (Cohen et al. 2000). Subsequent work showed that it was also activated by a wide variety of non-reading contrasts—picture naming versus viewing random lines, tactile discriminations of objects versus textures, rhyme vs pitch decisions, and so on (Price and Friston 2005, Table 1). The same pattern has recurred for all brain regions: the more neuroimaging that’s been done, the more functions each area seems to participate in.

In the other direction, individual cognitive functions can, at least sometimes, be performed by distinct sets of brain regions (Friston and Price 2003). Regions identified as crucial for some cognitive function on the basis of neuroimaging data, for example, can be lesioned with very little effect on that function. So the mapping between cognitive

functions and brain regions is many-many, and that’s bad for the standard reading.

1.2.2 Causal Context

No brain region does anything on its own: a bit of cortex in a petri dish will not read words, no matter how hard you try. Subtractive neuroimaging methods explicitly recognize this, and were designed to get around it. One *assumes*, say, that reading involves big swathes of brain; to get down to one specific bit, you match reading with a hypothetical control task in order to factor out that broad activation (along with the ever-present high baseline). Nobody thinks that this context is *irrelevant*, though: if you can’t see, you certainly can’t read. But then it seems that calling something a “region for reading” is problematic. No region reads on its own, in isolation from the rest.

The point is not *just* that causation always occurs against a background of other necessary conditions (though that is also a problem). Consider, for example, recently fashionable predictive coding models of brain function. Predictive coding models say that the primary goal of regions is to compare internally generated models with predictions fed back from higher levels of processing, and to pass on error signals that represent the difference. In such schemes, regional functioning simply can’t be understood without reference to the larger system in which it is embedded—for the prediction feedback from higher levels crucially determines what a region is computing (Price and Devlin 2011).

The brain may be thought of more like a machine shop than an assembly line. Each tool in a machine shop does a specific sort of thing. Each can do their specific sort of thing to a variety of different inputs, though, and participate in the creation of a variety of different products. What each tool does may not be obvious from the final product: what unifies table legs and birdhouses and gunstocks might be the need to mill a certain kind of groove, say. Making any product requires using a bunch of tools in the right sequence and order. There are also a variety of (more or less efficient) routes to the same end, using different tools in different sequences.

I find this a compelling picture of the brain. But it does make the ‘region for’ locution obscure. The PLF is differentially activated by both word reading and also a lot of other contrasts besides. What sense is there in calling it a “region for reading”?

1.2.3 Theory-Ladenness

Setting up subtractive experiments requires a control task to match against the experimental one. Determining a suitable control task requires assuming a particular cognitive theory: one which claims that looking at visually similar non-words differs from reading words by lacking a specialized word identification module (say). That’s what allowed us to identify R as the region for reading.

But of course, in any realistic scenario, differing cognitive theories will offer different accounts of what happens during a task. So one might, for example, have a theory on

which reading words differs only in *difficulty* from looking at non-words. Differential activation would then be evidence that R is the area that *works harder* in visual tasks. Note that some inferences actually work this way. Localizers for working memory, for example, look for increased activation in 2-back versus 1-back memory tasks. Nobody thinks that there's a new, distinct function that gets engaged by the 2-back task: there's one function that works harder in the 2-back condition.

So the standard story appears to give only *conditionals* of the form “If T then R is the region for f ,” where T is some background theory. That makes subsequent theory-testing tricky, of course. It's tempting to look at a scan and just say “Ah, R ! That does f , you know!” — that is, to strip off just the consequent, and not worry about whether the antecedent is actually consistent with what you're doing. In many cases it won't be, because the background theories we use often have implications in a variety of different domains. So there's a real danger of circularity here—not unavoidable, but very easy danger—when we assume T to get a functional attribution, and then later use that functional attribution to try to support T .

2 Brain Regions as Difference-Makers

2.1 An Analogy

The moral seems to be that the standard reading of “ R is a region for f ” is far too strong. Specificity is neither true nor helpful. Can we weaken it and get something more useful? I think we can.

Here, I think philosophy of psychology has hit upon a problem that philosophers of biology dispatched long ago. “ R is a region for f ” is roughly analogous to “ G is a gene for t ,” where t is some phenotypic trait. One finds the same problems: most genes code for many different phenotypic traits, many genes affect the same trait, genes only work against the causal background of the whole genome (and more besides), and so on. So it's very puzzling what to make of claims that thus-and-such is a *gene for* red eyes in *Drosophila*.

In the case of genes, however, a plausible solution to this puzzle has been offered. That's the idea that genes are *difference-makers* for traits. As Sterelny and Kitcher put it, talking about genes for eye color:

Nobody who subscribes to this practice of labelling believes that a pair of appropriately chosen stretches of DNA, cultured in splendid isolation, would produce a detached eye of the pertinent color. Rather, the intent is to indicate the effect that certain changes at a locus would make against the background of the rest of the genome. (Sterelny and Kitcher 1988, 349).

“ G is a gene for t ” means that varying the details of a gene at a particular locus would

lead to variation in t . In general: C is a difference-maker for E iff intervening on C would make a difference to E : that is, if we could step in and tweak the value of C directly, we'd be able to change the value of E by doing so. Difference-making picks out a relation of active counterfactual dependence: keep everything fixed and wiggle this part, and you'll get variations in this other part.

In the case of genes, thinking of genes as difference-makers is compatible with a gene being a difference-maker for other traits, and many things making a difference to that trait. Difference-makers are usually assumed to operate against a causal background, and don't compete with each other for causal influence.

2.2 Brain Regions

A similar move will help with brain regions. To say that " R is a region for f " is to say that that R , if manipulated, would have made a difference to whether or how a subject f -ed. This is a weaker relationship than posited by the standard picture, and so hopefully more plausible. Neuroscience and genetics have different concerns, of course, and so many details need to be hashed out. That will come shortly.

What do brain regions make a difference to, though? I'd suggested that f picks out cognitive-level functions on the standard model, but that's worth revisiting. There are three options for f .

First, f might pick out *neural*-level processes. It's surely true that brain bits make a difference to other brain bits, but also not very interesting for present purposes. I care mostly about testing cognitive theories, remember, and if we knew how to go from brain facts to cognitive theories we'd be in better shape than we are.

Second, f might pick out cognitive-level, subpersonal processes. Again, I think this is true but unhelpful: characterizing the difference-making relationship would require having a uncontroversial cognitive-level theory, which is again what we were actually looking for and so can't assume.

Third, f might pick out personal-level activities—reading, seeing, dancing, and other processes that are done by the whole person. This is neither obviously true nor obviously useful. I'm going to argue for both.

I'll mostly argue for utility, but a quick word about truth. I think treating f as personal-level is odd, but defensible. You might worry about crossing levels—that, for example, it somehow ends up confusing properties of parts for properties of wholes. But one of the advantage of the difference-making framework over ordinary causal relationships is that it allows this sort of synchronic difference-making relationship. Further, there's nothing intrinsically odd about crossing levels. Return to genes: they partly compose the organisms that they make a difference to, yet it makes perfect sense to talk about difference-making relationships. Many mechanistic explanations in neuroscience cross levels in precisely this way (? , Ch4)

2.3 The Formal Story

So I think treating f as personal-level is true (or at least plausible). Is it useful? Let's start by making it more precise.

To say that R makes a difference to a subject's f -ing (against a background of brain activity M) means that:

1. There is a set of ways of f -ing $\{a_1 \dots a_n\}$ which vary along some dimension (i.e. in the presence or the manner of f -ing)
2. There is a set of distinct patterns of R -activation $\{b_1 \dots b_n\}$,
3. When a subject f s in manner a_n , R has one of the patterns b_n ,
4. and intervening on R such as to change b_n to some other b_i while keeping (at least initially) M the same would have made the subject do one of the other a_i instead of a_n .

These four conditions extend the difference-making framework to brain regions. Condition one requires some meaningful sense in which the personal-level activity f can vary: reading can happen or not, but can also be fast or slow, of this word rather than that, and so on. No dimension of variation is privileged (though some will be more useful). Condition two ensures that there is a similarly well-defined sense in which regional activation can vary (assuming that it's activity in a region, rather than the region as such, that matters). Conditions three and four capture the idea that changing the way in which R was activated would make a difference to what happened with the subject's f -ing.

To flesh it out with an example: to say that the PLF is a region for word-reading is to say that the PLF is active in some way when you read words, and that if you altered that way you could change whether you read, or what you read, or how you read. That's what it means to make a difference to word-reading, and that's all it takes to be a region for word reading. This gets around the initial problems. It doesn't say that the PLF is specific to reading words (it could make a difference to other processes as well). It leaves open just *how* the PLF makes a difference. And it's compatible with many other things making a difference to reading words as well. Nevertheless, it does at least separate off PLF from things that don't affect reading, and that seems to be good reason to call it a region for reading words.

3 Ways of Making a Difference

3.1 Permissiveness

You might have noticed that my account is awfully permissive. A great many brain regions turn out to be difference-makers for word reading. Consider the reticular formation

(RF), a brainstem region that controls (among other things) sleep-wake cycles. RF is a difference-maker for word reading: change its activation in the right way and you go from reading words to not (because you're not conscious). Or take an early visual area like V1. It can make a difference like RF—remove activation and you'll be blind, which means you can't read. But V1 also contains roughly retinotopic representations of what you see; it's a way-station on the way to more complex visual processing. So when you read, we could have altered your V1 activity in such a way that you would have read something different.

On my account, then, both RF and V1 end up regions for word-reading too. I think that's still *right*—being conscious does make a difference to whether you read, as do facts about what you see—but you might wonder how *useful* it is to point out.

Here, I think the difference-maker framework provides a nice move. As Woodward notes, most philosophical work on causation has focused on what makes something a causal relationship at all. But there is another important project, that of “elucidating and understanding the basis for various distinctions that we (both ordinary folk and scientists) make *among* casual relationships.” ((Woodward 2010) 287). More generally, once we move to thinking about difference-makers, it becomes important to think about *how* something makes a difference.

With that in mind, I'll detail two aspects of difference-making that distinguish PLF as a more interesting region than RF or V1.

3.2 Specificity

A first distinction among causes is *specificity*, in Woodward's ‘influence’ sense. A more specific cause is one that has more fine-grained influence over the effect. In general, a specific relationship between C and E is one where, by manipulating C into many different states, we can reliably manipulate E into many different states. As Woodward puts it, “ C will influence E to the extent that by varying the state of C and its time and place of occurrence, we can modulate the state of E in a fine-grained way” (Woodward 2010, 305).

Specificity is a graded notion. At one end, we have what Woodward calls ‘switch-like’ influence. C has switch-like influence over E when E can only be manipulated into one of two states. In Woodward's well-known radio analogy, the on-off switch of a radio gives switch-like influence over the tuned station. No matter how many effective positions there are of the switch, the switch can only flip the radio between two states (off, and whatever station it happens to be tuned to). At the other end is ‘dial-like’ influence. C has dial-like influence over E just in case C can be in many states and there there is a 1-to-1 relation between states of C and states of E . In Woodward's radio analogy, the tuning knob has dial-like influence over the station: each stop along the dial corresponds (more or less) to a different station.

In general, dial-like influence is more interesting and more useful. Lots of things tend

to have switch-like influence over a phenomenon of interest, including most background conditions. That's true for the brain as well. Manipulations of RF, and crude manipulations of V1, are difference-makers for reading, but not in any distinctive way. At best, they tell us that we need to be conscious, or able to see, to read. We knew that already. PLF, by contrast, might have more fine-grained influence over reading: different states of PLF might correspond to distinct words (say), and so we can vary not just whether you read but what you read and how you read by varying PLF.

3.3 Systematicity

Specificity alone won't do the trick, however. Consider: since V1 roughly replicates early visual information, each difference in a seen word should correspond to some difference in V1 activation. Hence more subtle manipulations of V1 may give us specific influence over which word you read.

Let's introduce a second distinction, inspired again by Woodward's radio. Some things make more *systematic* differences than others. Suppose I have a radio with an antenna. Suppose further that there is a relationship between signal quality and antenna position, but that relationship is (as it often is) maddeningly obscure. Move the antenna one degree to the left and the signal gets clearer, another degree to the left and it gets drastically worse, another degree makes it just a tad worse, and so on.

Both the antenna and the tuning dial may have a fine-grained relationship to signal strength and station, respectively. That is, each position of the antenna might correspond to a distinct signal-to-noise ratio, and vice versa. Nevertheless, there's an important difference between the two. Only the dial has an *systematic* relationship between its states and what it controls. An systematic relationship is one where if you're in one state of E , it's obvious how to get to any other state of E , and it's obvious how a change in C will affect E , and this is obvious even if you don't know the relationship ahead of time.

That's a little loose, but I think it can be made more precise. Represent possible variations in C in a (possibly multi-dimensional) state-space. Then a relationship between C and E is *systematic* just in case the different states of C that give rise to the same E are located in contiguous regions of that state-space, and states that give rise to similar states of E are located closer together than states that give rise to very different states of E .

The tuning dial on the radio counts as systematic in this sense: there might be a bit of play in the dial, but generally states that give rise to the same station are all located together on the dial. Further, the closer two stations are, the more similar their position on the dial. Unsystematic relationships, by contrast, have states of C that correspond to the same E located in far-flung parts of that state-space, or else have states that give rise to similar E s further apart than states that give rise to very different E s. The antenna's relationship to signal quality is unsystematic in this sense.

Systematicity ought to be a graded notion. In general, systematicity will be a function of the average distance between C s that give similar effects and the average distance

between the nearby *Es* vs distant *Es*.

Systematicity can't always be attained: sometimes the word is just unsystematic. Systematicity does not appear to play a big role in thinking about the relationship between genotype and phenotypes. But if systematicity *is* in place, it gives us a very useful set of principles by which to manipulate the variables we care about. That's crucial on difference-making accounts of explanation, in the same way that specificity is crucial. For again, in difference-making we care about the degree of control a variable gives us over a system, and systematic difference-making gives us much better control.

Further, systematicity seems to be both present and theoretically important in the brain. There are a number of spatially organized maps in the brain: retinotopy in V1, somatotopy in primary sensory cortex, and so on. Given that, I think there might be a straightforward sense in which the PLF has a more systematic relationship to word-reading than V1. V1, remember, is roughly retinotopic. The same word can appear in a variety of places in the visual field, in a variety of fonts and orientations, and so on. This makes the relationship between V1 and words relatively unsystematic (even if it is specific): activation patterns representing different words are close together in V1-space, while patterns representing the same word can be far apart. PLF—or *some* region, at least—takes this complexity and reduces it down to a more orderly code, one that allows us to read the same word no matter where or how it appears.

Systematicity and specificity together thus give us an intuitive way to carve up difference-making in the brain. PLF, V1, and the reticular formation are all regions for reading. But within regions for reading, we can distinguish between those that give us insensitive or disorderly manipulable relationships and those that give fine-grained, orderly ones. Reading is just an example (and a vaguely sketched one at that.) The same sort of story can be applied to any activity, and to any brain region that makes a difference to that activity.

4 The Upshot

Treating brain regions as difference-makers solves the metaphysical problems that plagued the standard story. But of course, I'm also interested in theory-testing. I conclude with some reflections on how this framework might be applied to the methodological problems of theory-testing with neuroimaging.

Here's a very simple version. Neuroimaging gives us information about the difference-makers for the tasks we care about. It's not a perfect evidence, and it comes with many caveats, but it's at least plausible that neuroimaging does so. Discovering (say) that PLF is active for viewing words versus non-words is evidence that PLF makes a difference to word reading. Using neuroimaging also gets around some of the more *recherché* metaphysical worries, as brain imaging gives us evidence for the *actual* difference-makers between tasks (Waters 2007). Further, since this doesn't commit you to anything about which cognitive theory is true—for the description of difference-making remains at the task level—so it doesn't have the obvious theory-dependence that plagued the standard

account.

On the down side, it doesn't commit you to anything about which cognitive theory is true. So it looks like there's nothing that weighs for or against any cognitive theory, which is what we wanted to happen. (The fact that there's *some* difference maker is of course unsurprising, whichever theory is true.)

Here, though, I think there's something to say. I've spoken of personal-level activities in a loose and informal way. Part of the job of psychology, however, is to systematize those personal-level theories into a taxonomy of activities. That taxonomy is important because it forms the groundwork for cognitive theories. As cognitive theories predict and explain the structure of personal-level tasks, discovering facts about the structure of personal-level tasks can weigh for or against cognitive theories.

Here, I think there's an analogy with neuropsychology, at least on the classic Shallice & Coltheart model (Shallice 1988; Coltheart 2001). Lesioned patients give us evidence about cognition. How? Not by some direct link between brain facts and cognitive facts. Rather, lesion evidence structures the taxonomy of personal-level tasks—showing, in the classic case, that performance on two tasks can be completely dissociated. This indirectly constraining cognitive theories, by restricting the true theories to the ones that can adequately account for the dissociation.

I think there's a similar possibility for neuroimaging. The rough idea is that similar tasks should vary along more and more closely related dimensions than distinct tasks. So if we have evidence of fine-grained, systematic difference-making among (say) word-reading but we can only get relatively crude switch-like difference-making between faces and houses, then we have evidence that they're distinct tasks. That constrains cognitive theories, since they have to explain how that's possible.

Do we actually *have* evidence of this? That's a complex question. The simple version is that most experiments don't give us what we need. Simple subtractions, even lots of them, usually only give evidence for switch-like difference-making. That's why, I suspect, they rarely settle the question between competing theories. But more complex data analyses might have a better shot. Techniques like multi-voxel pattern analysis give evidence for systematicity, while rigorous meta-analyses give evidence for systematic convergence and divergence.

Of course, the devil will be in the details. I think the details are worth sorting out, though. Switching to a difference-making account gives us a sense of brain regions that fits with empirical data. It also gives us a vocabulary with which we can talk about more fine-grained differences and similarities between regions. And those fine-grained orderly differences might, at a first pass, give us a better clue to what's going on in both the brain and the mind.

xxx words

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